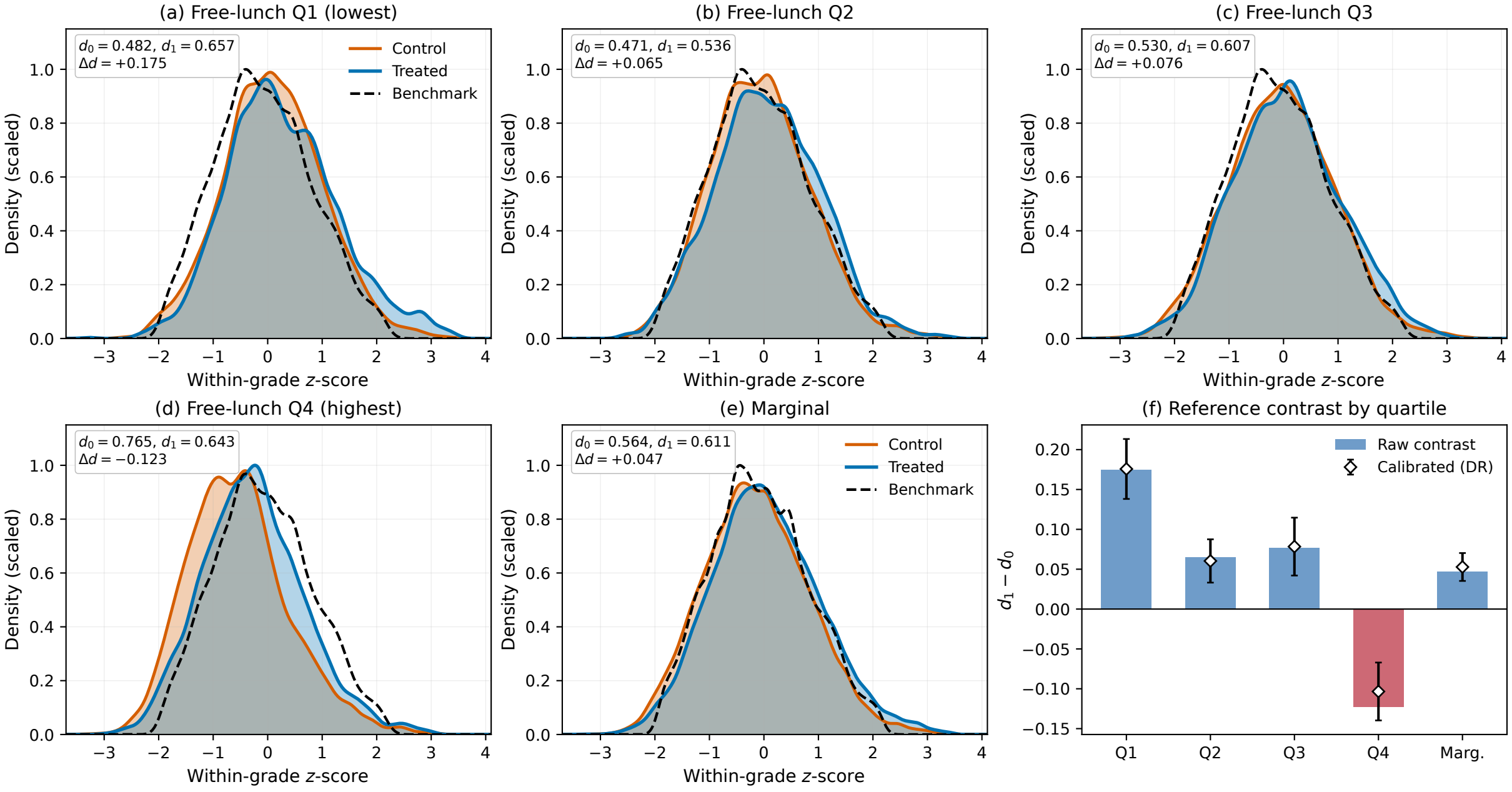

(a) Free-lunch Q1 (lowest)
$d_0 = 0.482$, $d_1 = 0.657$
$\Delta d = +0.175$
Control
Treated
Benchmark
(b) Free-lunch Q2
$d_0 = 0.471$, $d_1 = 0.536$
$\Delta d = +0.065$
(c) Free-lunch Q3
$d_0 = 0.530$, $d_1 = 0.607$
$\Delta d = +0.076$
(d) Free-lunch Q4 (highest)
$d_0 = 0.765$, $d_1 = 0.643$
$\Delta d = -0.123$
(e) Marginal
$d_0 = 0.564$, $d_1 = 0.611$
$\Delta d = +0.047$
(f) Reference contrast by quartile
Raw contrast
Calibrated (DR)
$d_1 - d_0$
Q1
Q2
Q3
Q4
Marg.
Density (scaled)
Within-grade z-score